SAGAR PUBLIC SCHOOL

Integrated Programme • Topic: Current Electricity

Current Electricity – Level 1

Name: _____ **Class / Sec:** _____ **Date:** _____

Time Allowed: 60 Mins • Total Questions: 20 • JEE Main Previous Year Questions

A Ammeter & Voltmeter Conversion (Q1–5) **B** Voltmeter Loading & Connections (Q6–8) **C** Galvanometer (Q9–14) **D** Ammeter Reading & Potentiometer (Q15–20)

SECTION A – Ammeter & Voltmeter Conversion (Q1–Q5)

Q1. *[PYQ · JEE Main 2026, 24 Jan Evening]*

A moving coil galvanometer of resistance 100 Ω shows a full scale deflection for a current of 1 mA. The value of resistance required to convert this galvanometer into an ammeter, showing full scale deflection for a current of 5 mA, is _____ Ω.

(A) 25 (B) 2.5 (C) 0.5 (D) 10

Q2. *[PYQ · JEE Main 2025, 23 Jan Evening]*

A galvanometer having a coil of resistance 30 Ω need 20 mA of current for full-scale deflection. If a maximum current of 3 A is to be measured using this galvanometer, the resistance of the shunt to be added to the galvanometer should be $\frac{30}{X}$ Ω, where X is

(A) 447 (B) 298 (C) 149 (D) 596

Q3. *[PYQ · JEE Main 2024, 1 Feb Evening]*

In an ammeter, 5% of the main current passes through the galvanometer. If resistance of the galvanometer is G, the resistance of ammeter will be:

(A) $199\,G$ (B) $200\,G$ (C) $\frac{G}{20}$ (D) $\frac{G}{199}$

Q4. *[PYQ · JEE Main 2024, 1 Feb Morning]*

A galvanometer has a resistance of 50 Ω and it allows maximum current of 5 mA. It can be converted into voltmeter to measure up to 100 V by connecting in series a resistor of resistance:

(A) 19500 Ω (B) 5975 Ω
(C) 20050 Ω (D) 19950 Ω

Q5. *[PYQ · JEE Main 2020, 5 Sep Morning]*

A galvanometer of resistance G is converted into a voltmeter of range 0–1 V by connecting a resistance R_1 in series with it. The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to 0–2 V will be:

(A) G (B) R_1 (C) $R_1 + G$ (D) $R_1 - G$

SECTION B – Voltmeter Loading & Connections (Q6–Q8)

Q6. *[PYQ · JEE Main 2026, 24 Jan Morning]*

Two resistors of 100 Ω each are connected in series with a 9 V battery. A voltmeter of 400 Ω resistance is connected to measure the voltage drop across one of the resistors. The voltmeter reading is _____ V.

(A) 3 (B) 2 (C) 4 (D) 4.5

Q7. *[PYQ · JEE Main 2020, 3 Sep Evening]*

Two resistors 400 Ω and 800 Ω are connected in series across a 6 V battery. The potential difference measured by a voltmeter of 10 kΩ across 400 Ω resistor is close to:

(A) 2.05 V (B) 1.95 V (C) 2 V (D) 1.8 V

Q8. *[PYQ · JEE Main 2020, 6 Sep Evening]*

A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit:

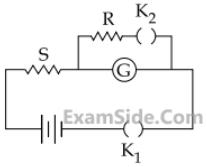
(A) Ammeter is always connected in series and voltmeter in parallel
(B) Both ammeter and voltmeter must be connected in series
(C) Both ammeter and voltmeter must be connected in parallel
(D) Ammeter is always used in parallel and voltmeter is series

SECTION C – Galvanometer (Q9–Q14)

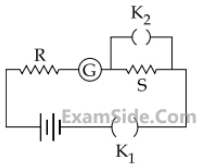
Q9. *[PYQ · JEE Main 2014, 11 Apr Morning]*

In the circuit diagrams (A, B, C and D) shown below, R is a high resistance and S is a resistance of the order of galvanometer resistance G. The correct circuit, corresponding to the half deflection method for finding the resistance and figure of merit of galvanometer, is the circuit labelled as:

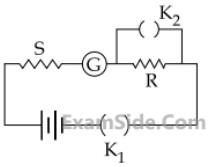
(A)



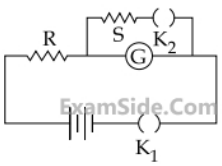
(B)



(C)



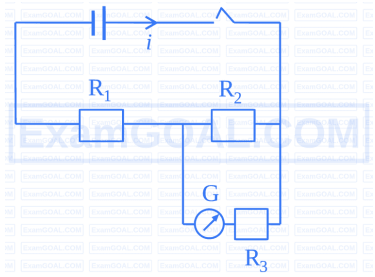
(D)



Q10. *[PYQ · JEE Main 2013, 22 Apr Morning]*

To find the resistance of a galvanometer by the half deflection method, the following circuit is used with resistances $R_1 = 9970\,\Omega$, $R_2 = 30\,\Omega$ and $R_3 = 0$. The de-

flection in the galvanometer is d. With $R_3 = 107\,\Omega$, the deflection changed to $\frac{d}{2}$. The galvanometer resistance is approximately:



(A) $107\,\Omega$ (B) $137\,\Omega$ (C) $\frac{107}{2}\,\Omega$ (D) $77\,\Omega$

Q11. *[PYQ · JEE Main 2020, 5 Sep Evening]*

A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of 6 mA it produces a deflection of 2°, its figure of merit is close to:

(A) 6×10^{-3} A/div (B) 666° A/div
(C) 3×10^{-3} A/div (D) 333° A/div

Q12. *[PYQ · JEE Main 2021, 27 Aug Evening]*

For full scale deflection of total 50 divisions, 50 mV voltage is required in galvanometer. The resistance of galvanometer if its current sensitivity is 2 div/mA will be:

(A) $1\,\Omega$ (B) $5\,\Omega$ (C) $4\,\Omega$ (D) $2\,\Omega$

Q13. *[PYQ · JEE Main 2023, 11 Apr Morning]*

The current sensitivity of moving coil galvanometer is increased by 25%. This increase is achieved only by changing in the number of turns of coils and area of cross section of the wire while keeping the resistance of galvanometer coil constant. The percentage change in the voltage sensitivity will be:

(A) +25% (B) −50% (C) −25% (D) Zero

Q14. *[PYQ · JEE Main 2025, 23 Jan Morning]*

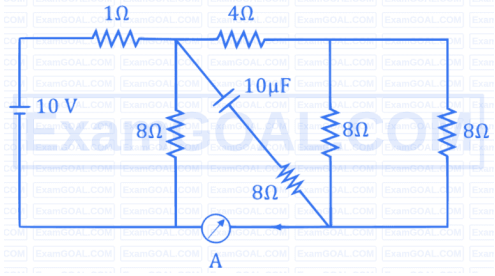
Consider a moving coil galvanometer (MCG):
A. The torsional constant in moving coil galvanometer has dimensions $[ML^2T^{-2}]$.
B. Increasing the current sensitivity may not necessarily increase the voltage sensitivity.
C. If we increase number of turns (N) to its double ($2N$), then the voltage sensitivity doubles.
D. MCG can be converted into an ammeter by introducing a shunt resistance of large value in parallel with galvanometer.
E. Current sensitivity of MCG depends inversely on number of turns of coil.
Choose the correct answer from the options given below:

(A) A, D Only (B) A, B Only
(C) A, B, E Only (D) B, D, E Only

SECTION D – Ammeter Reading & Potentiometer (Q15–Q20)

Q15. *[PYQ · JEE Main 2026, 24 Jan Evening]*

The reading of the ammeter (A) in steady state in the following circuit (assuming negligible internal resistance of the ammeter) is _____ A.



(A) 1/2 (B) 1 (C) 2 (D) 0

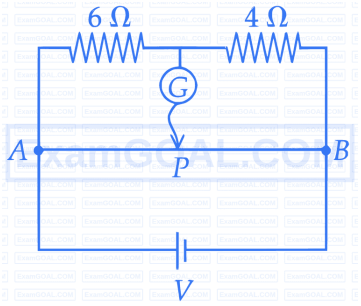
Q16. *[PYQ · JEE Main 2026, 23 Jan Evening]*

To compare EMF of two cells using potentiometer the balancing lengths obtained are 200 cm and 150 cm. The least count of scale is 1 cm. The percentage error in the ratio of EMFs is _____

(A) 1.75 (B) 1.55 (C) 1.17 (D) 1.65

Q17. *[PYQ · JEE Main 2026, 21 Jan Evening]*

The total length of potentiometer wire AB is 50 cm in the arrangement as shown in the figure. If P is the point where the galvanometer shows zero reading then the length AP is _____ cm.



(A) 25 (B) 30 (C) 15 (D) 20

Q18. *[PYQ · JEE Main 2024, 4 Apr Morning]*

To measure the internal resistance of a battery, potentiometer is used. For $R = 10\,\Omega$, the balance point is observed at $l = 500$ cm and for $R = 1\,\Omega$ the balance point is observed at $l = 400$ cm. The internal resistance of the battery is approximately:

(A) $0.1\,\Omega$ (B) $0.3\,\Omega$ (C) $0.2\,\Omega$ (D) $0.4\,\Omega$

Q19. *[PYQ · JEE Main 2023, 29 Jan Evening]*

A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by $5\,\Omega$. When a resistance of $15\,\Omega$ is used for shunting, null point moves to 300 cm. The internal resistance of the cell is _____ Ω.

Q20. *[PYQ · JEE Main 2014, 12 Apr Morning]*
In the experiment of calibration of voltmeter, a standard cell of e.m.f. 1.1 volt is balanced against 440 cm of potentiometer wire. The potential difference across the ends of resistance is found to balance against 220 cm of the wire. The corresponding reading of voltmeter is 0.5 volt. The error in the reading of the voltmeter will be:

(A) −0.15 volt (B) 0.15 volt
(C) 0.5 volt (D) −0.05 volt

SELF ASSESSMENT (to be filled by the student after the test)

Time Taken: _____ min **Attempted:** _____ / 20 **Unattempted:** _____

Correct: _____ **Incorrect:** _____ **Accuracy:** _____ %

Section	Questions	Correct	Incorrect	Topics / concepts I must revise
A – Ammeter & Voltmeter Conversion	5			
B – Voltmeter Loading & Connections	3			
C – Galvanometer	6			
D – Ammeter Reading & Potentiometer	6			

Personal Feedback (what went well / what went wrong / silly mistakes / plan for next sheet):

_____

_____

_____

_____

Student's Signature: _____ **Teacher's Remark / Signature:** _____